PHASE 3 / STRATEGY RESEARCH

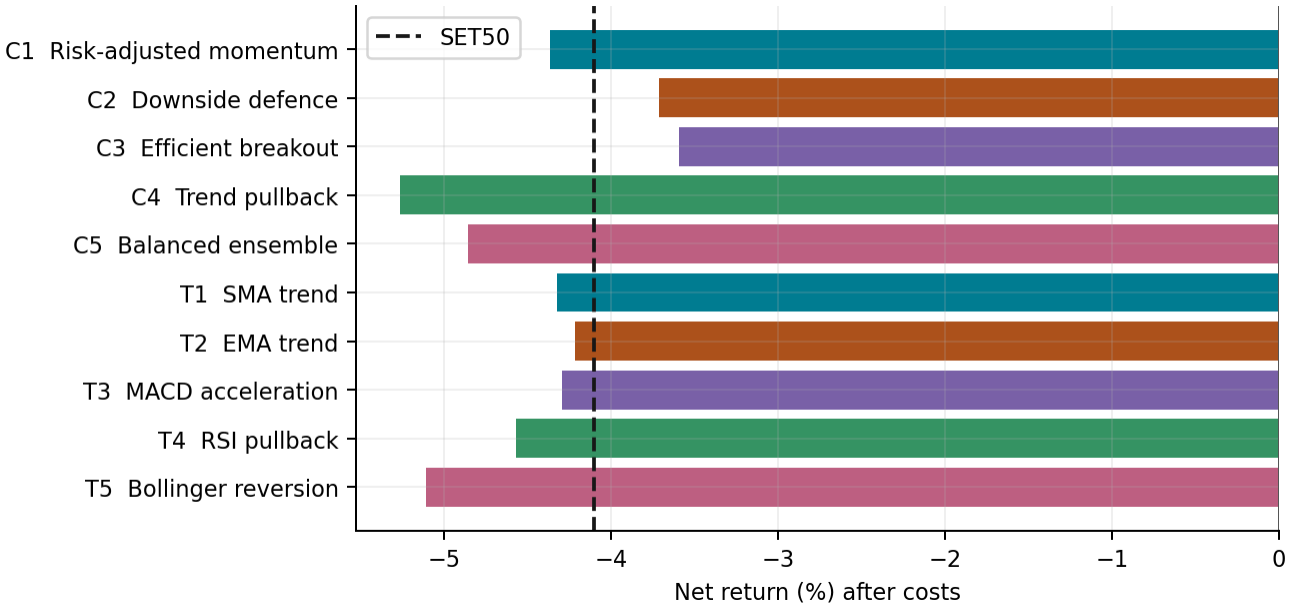
Ten strategies. One controlled comparison.

KGI CU league · 30 October 2026 · TradeSim 0.2.7

0/10 strategies pass the measured common criteria; 0/10 pass the Best Alpha checks; 10/10 meet the requested 70–80% average exposure band. The development lead under the declared ranking is C3 (Efficient breakout): net return -3.590%, excess +0.510 pp, drawdown 4.310%, average exposure 74.154%.

No prize-ready winner is established. All ten hypotheses were fixed before this comparison, but September was already inspected in earlier work. The ranking is development evidence only; no parameter search or retrospective change was made after these results. The original alphabetical default remains the reproducible engineering reference; a suite strategy runs through its explicit configuration.

The original reference returned -2.992% at 69.694% average exposure. The new-suite lead differs by -0.598 pp of net return. Different baskets and exposure mean this is a practical portfolio comparison, not an isolated signal effect.



Read this report as development evidence. Phase 3 engineering is complete; profitability and Phase 4 validation are separate milestones.

Results at a glance

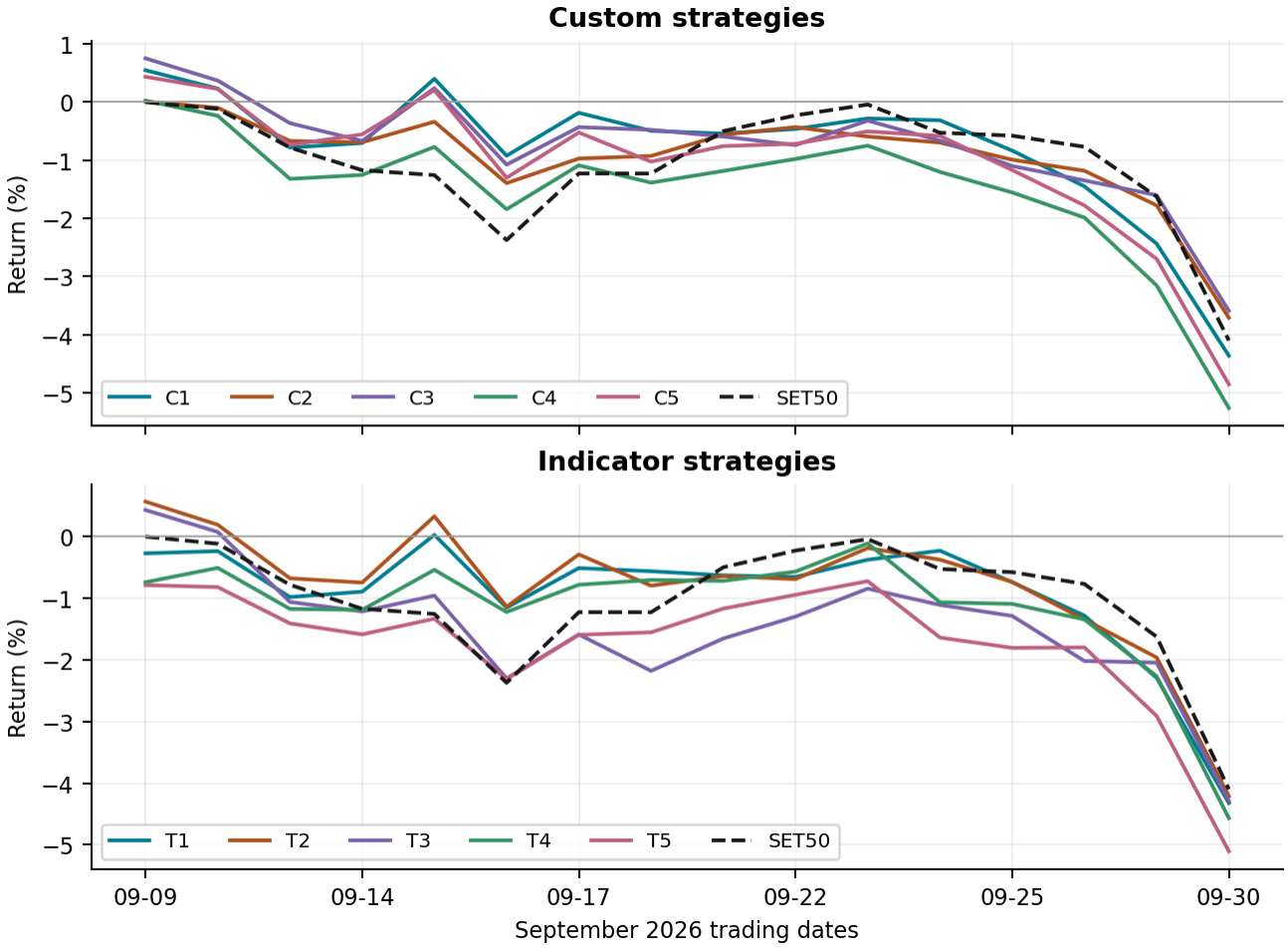
SET50 returned -4.099496% and its maximum drawdown was 4.099496%. The strict common drawdown boundary is therefore 6.149244%. Statuses use unrounded values. Drawdowns are positive loss magnitudes; return and excess retain their signs.

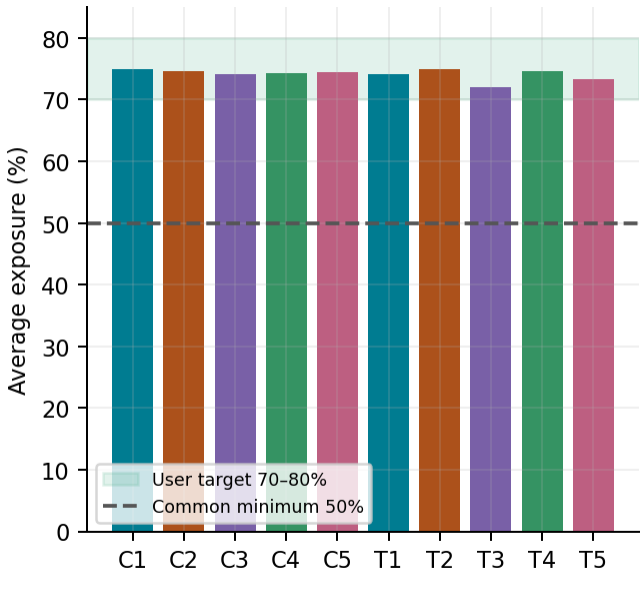
ID	Net %	SET50 %	Excess pp	MaxDD %	SET50 MaxDD %	Avg exposure %
C1	-4.362	-4.099	-0.262	4.884	4.099	74.869
C2	-3.710	-4.099	0.389	3.718	4.099	74.668
C3	-3.590	-4.099	0.510	4.310	4.099	74.154
C4	-5.263	-4.099	-1.163	5.288	4.099	74.254
C5	-4.855	-4.099	-0.756	5.268	4.099	74.396
T1	-4.324	-4.099	-0.224	4.350	4.099	74.123
T2	-4.213	-4.099	-0.113	4.754	4.099	74.899
T3	-4.290	-4.099	-0.190	4.701	4.099	72.087
T4	-4.570	-4.099	-0.470	4.464	4.099	74.689
T5	-5.107	-4.099	-1.008	4.416	4.099	73.351

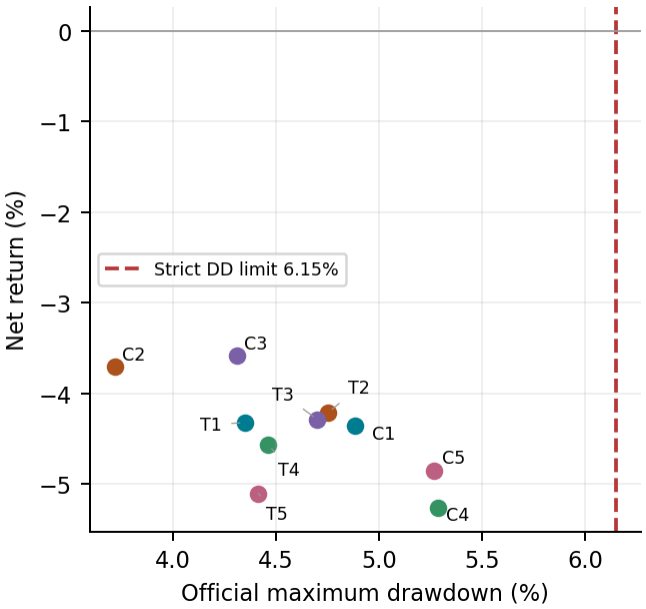
ID	Stocks	FIFO lots	Payoff THB	Fees THB	Common	Best Alpha	70–80%
C1	10	25	-1,125.90	17,949.69	fail	fail	yes
C2	10	5	-977.95	15,459.78	fail	fail	yes
C3	10	34	-603.88	19,905.19	fail	fail	yes
C4	10	39	-360.27	18,002.79	fail	fail	yes
C5	10	39	-825.33	19,214.41	fail	fail	yes
T1	10	17	-472.03	15,515.93	fail	fail	yes
T2	10	25	-393.96	16,227.38	fail	fail	yes
T3	10	61	-618.61	23,953.53	fail	fail	yes
T4	10	54	-654.89	26,261.28	fail	fail	yes
T5	10	72	-2,221.38	33,162.82	fail	fail	yes

Expected payoff is THB per completed non-breakeven FIFO lot; it is not total portfolio profit. Common and Best Alpha status are local checks under the working rules. Every run trades exactly ten distinct stocks.

Returns, drawdown and exposure







Return charts use initial capital for strategies and the first sample close for SET50, matching official reporting. The first-day anchors differ; aligned first-close returns are available in the machine-readable evidence. Dates in charts are month-day, September 2026.

Risk controls and implementation

One shared execution layer: THB 10 million, unchanged TradeSim 0.2.7, 16 supplied tick days (9–30 September 2026), 2,906,167 ticks per full run. All 48 supplied symbols are replayed in the original notebook symbol/row-index order, not globally sorted time.

Before September 9, use at most 180 prior daily bars and require at least 80 valid observations. Form the 30-stock pool with the highest median 20-session close × volume. Each signal selects ten stocks, at most two per sector. Freeze those names for the whole run; daily re-ranking changes weights only.

Normal target exposure is 76% of opening NAV. Ranked weights range from 9.12% to 6.08% per stock. Buy checks limit each position to 10%, total investment to 80%, and preserve 20% cash. Later price movements can exceed these pre-buy limits.

Opening drawdown guard lowers target to 72% at 2% portfolio drawdown or 1.2× SET50 drawdown when SET50 drawdown is at least 0.5%. It lowers target to 55% at 4% portfolio drawdown or 1.35× SET50 drawdown when SET50 drawdown is at least 1%. The guard uses drawdown observed so far, including initial capital for the portfolio, and does not reset a historical maximum after recovery.

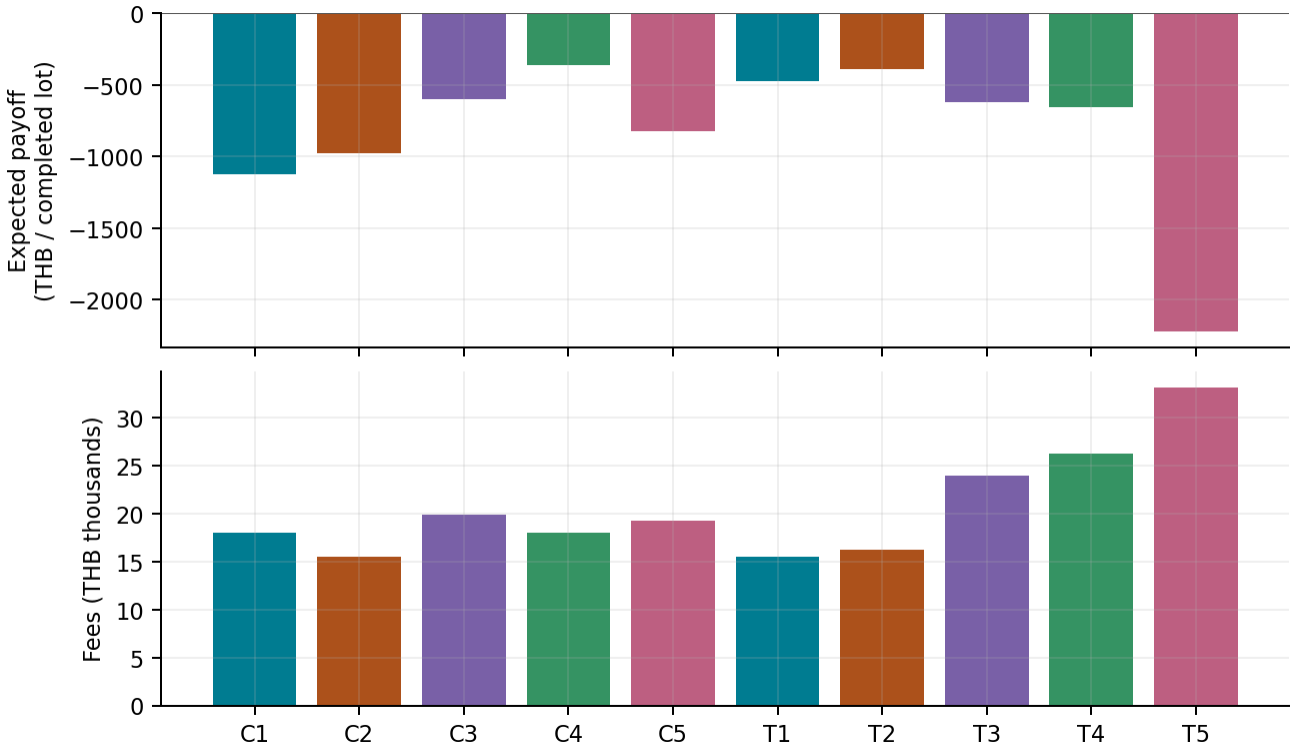
Quantity targets use prior close and 100-share lots. Rebalance when the opening gap is at least 0.4% of NAV or the position is absent. Retry actual residual quantities at most once per symbol per 15-minute decision slot. Market IOC fills may be partial or zero. Costs are 0.157% commission plus 7% VAT on commission per side, and the official adverse price tick.

The drawdown guard is evaluated daily and remains subject to gaps and incomplete fills. It can reduce average exposure below the user’s 70–80% preference; that tradeoff is reported. There is no forced end-of-sample liquidation, cosmetic trade generation, or guarantee of eligibility.

These are daily signal-ranked allocations, not binary crossover systems. The same risk layer makes comparisons clearer, but does not isolate signal effects from basket selection. Sector diversification is limited to two names per sector; it is not an explicit correlation cap.

Cash, holdings, raw open costs, NAV, realized P&L, fees, FIFO completion, payoff and benchmark statistics reconcile independently with the official engine every day. No engine or source market-data file was modified.

Payoff, friction and earlier experiments



The earlier reference used an alphabetical basket and roughly 70% initial allocation. Its two exit experiments used net +2% profit-taking, −4% loss exits and ten-session maximum holding age; the second also used 20-session momentum. They are historical context, not controlled comparisons with the new 76% allocation models.

Earlier run	Net %	Exposure %	FIFO lots	Payoff THB	Common
Original reference	-2.992	69.694	0	Undefined	fail
Alphabetical + exits	-4.228	57.033	349	-999.31	fail
Momentum + exits	-5.551	54.948	378	-1,390.41	fail

Completing more lots alone did not improve the earlier strategies: both exit variants increased fees and had negative payoff. The current suite uses allocation rebalancing to create economically motivated trades; it never manufactures a round-trip count.

C1 · Risk-adjusted momentum

CUSTOM STRATEGY

Favor sustained gains that are large relative to day-to-day volatility. Blend the shorter trend with the longer trend to avoid relying on one lookback.

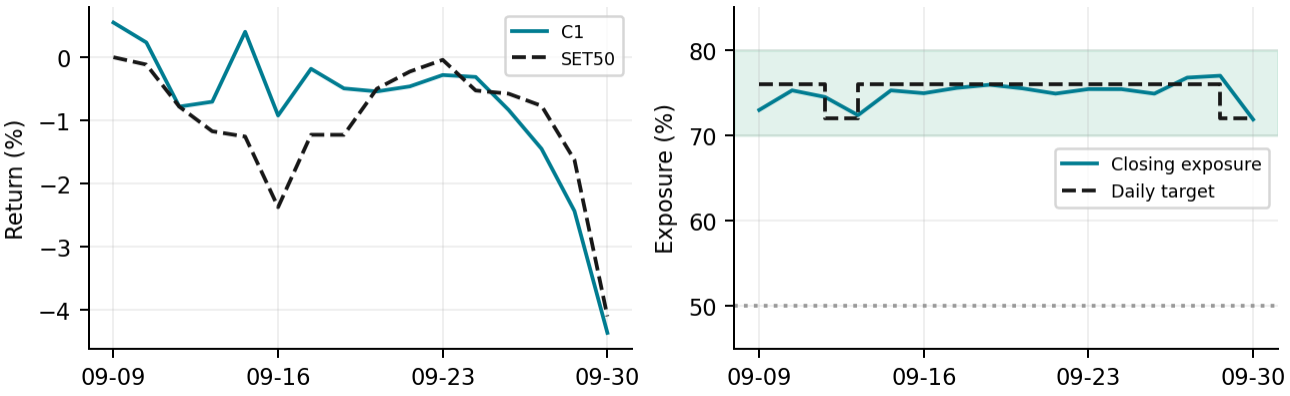
Score: (0.6 × return20 + 0.4 × return60) / (volatility20 × √20)

Inputs: 20- and 60-session returns; 20-session return volatility.

Frozen basket and initial target weights: BCP (9.12%), PTTGC (8.78%), KKP (8.44%), BBL (8.11%), PTT (7.77%), CPF (7.43%), IVL (7.09%), KTC (6.76%), CRC (6.42%), BH (6.08%)

Metric	Value	Metric	Value
Net return	-4.362%	SET50 return	-4.099%
Excess return	-0.262 pp	SET50 MaxDD	4.099%
Official MaxDD	4.884%	Initial-capital MaxDD	4.884%
Average exposure	74.869%	Final exposure	71.826%
Distinct stocks	10	Completed FIFO lots	25
Expected payoff	THB -1,125.90	Fees	THB 17,949.69
Realized P&L	THB -43,040.04	Unrealized P&L	THB -393,149.65

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 14 days, 72% target on 2 days.



Risk to this hypothesis: Momentum can reverse abruptly; low recent volatility does not imply low future risk.

Largest closing position: 9.39%. Maximum measured daily replay time: 26.57s. Timing is local and includes concurrent experiments; it is not a production guarantee.

C2 · Downside defence

CUSTOM STRATEGY

Favor stocks with smaller downside fluctuations and less positive sensitivity to SET50. This is a defensive allocation hypothesis, not a direct forecast of rising prices.

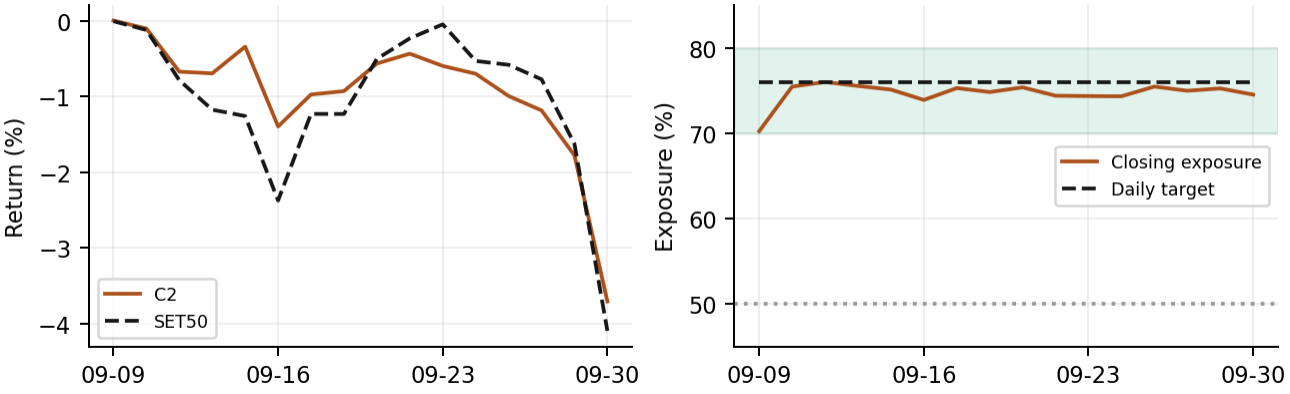
Score: −(annualized downside deviation20 + 0.1 × max(beta60, 0))

Inputs: Negative daily returns over 20 sessions; up to 60 aligned stock/SET50 return intervals.

Frozen basket and initial target weights: PTT (9.12%), PTTEP (8.78%), SCB (8.44%), BBL (8.11%), BDMS (7.77%), CPF (7.43%), MINT (7.09%), TRUE (6.76%), IVL (6.42%), SCC (6.08%)

Metric	Value	Metric	Value
Net return	-3.710%	SET50 return	-4.099%
Excess return	0.389 pp	SET50 MaxDD	4.099%
Official MaxDD	3.718%	Initial-capital MaxDD	3.718%
Average exposure	74.668%	Final exposure	74.503%
Distinct stocks	10	Completed FIFO lots	5
Expected payoff	THB -977.95	Fees	THB 15,459.78
Realized P&L	THB -14,365.47	Unrealized P&L	THB -356,634.30

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 16 days.



Risk to this hypothesis: Low downside risk can lag a rebound. Beta is estimated from a short and changing sample.

Largest closing position: 9.17%. Maximum measured daily replay time: 27.02s. Timing is local and includes concurrent experiments; it is not a production guarantee.

C3 · Efficient breakout

CUSTOM STRATEGY

Favor prices near their previous 55-session high, especially when the path there was relatively direct. The efficiency term rewards persistent movement.

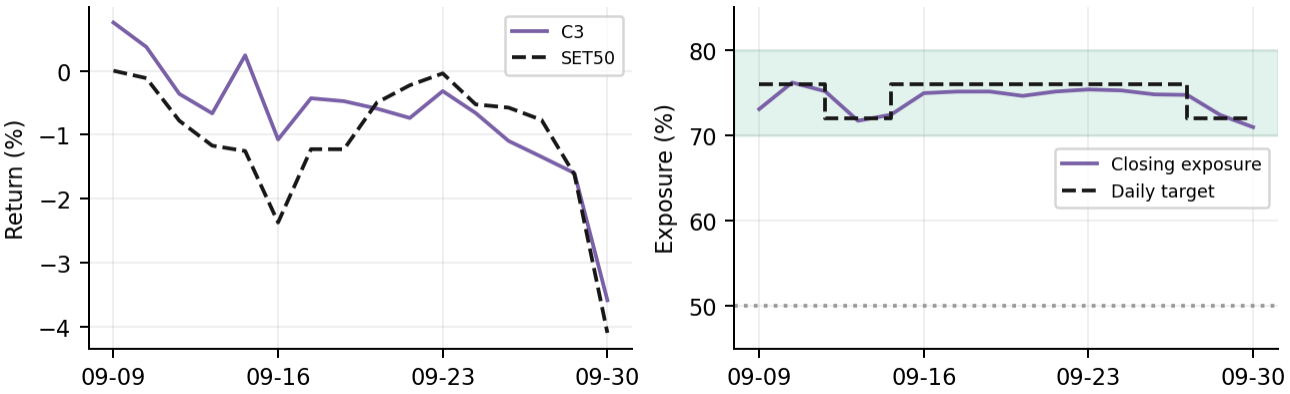
Score: (close / prior high55 − 1) / volatility20 + 0.5 × efficiency20

Inputs: Previous 55-session high excluding the latest bar; 20-session volatility and path efficiency.

Frozen basket and initial target weights: IVL (9.12%), PTTGC (8.78%), PTT (8.44%), SCC (8.11%), PTTEP (7.77%), CPF (7.43%), CCET (7.09%), KKP (6.76%), CRC (6.42%), BH (6.08%)

Metric	Value	Metric	Value
Net return	-3.590%	SET50 return	-4.099%
Excess return	0.510 pp	SET50 MaxDD	4.099%
Official MaxDD	4.310%	Initial-capital MaxDD	4.310%
Average exposure	74.154%	Final exposure	70.925%
Distinct stocks	10	Completed FIFO lots	34
Expected payoff	THB -603.88	Fees	THB 19,905.19
Realized P&L	THB -25,478.24	Unrealized P&L	THB -333,471.95

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 12 days, 72% target on 4 days.



Risk to this hypothesis: False breakouts and gaps can create losses before a daily risk guard reacts.

Largest closing position: 10.07%. Maximum measured daily replay time: 30.92s. Timing is local and includes concurrent experiments; it is not a production guarantee.

C4 · Trend pullback

CUSTOM STRATEGY

Favor a strong longer trend with a recent pullback. The recent return is subtracted to distinguish a potential recovery entry from a stretched move.

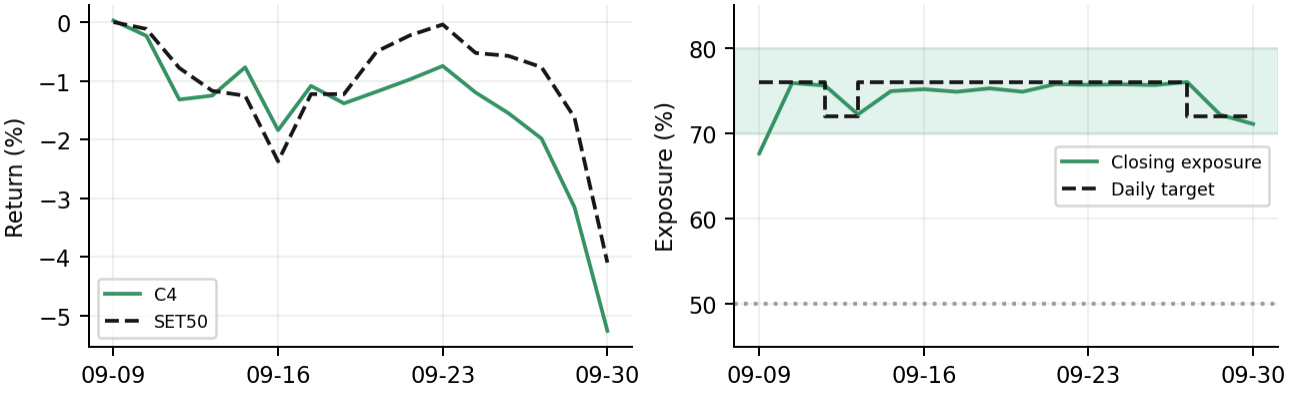
Score: (return60 − 2 × return5) / (volatility20 × √20)

Inputs: 60-session trend, 5-session pullback, and 20-session volatility.

Frozen basket and initial target weights: KBANK (9.12%), BBL (8.78%), BCP (8.44%), KTC (8.11%), GPSC (7.77%), CPF (7.43%), CRC (7.09%), PTTGC (6.76%), MTC (6.42%), BDMS (6.08%)

Metric	Value	Metric	Value
Net return	-5.263%	SET50 return	-4.099%
Excess return	-1.163 pp	SET50 MaxDD	4.099%
Official MaxDD	5.288%	Initial-capital MaxDD	5.288%
Average exposure	74.254%	Final exposure	71.064%
Distinct stocks	10	Completed FIFO lots	39
Expected payoff	THB -360.27	Fees	THB 18,002.79
Realized P&L	THB -32,288.52	Unrealized P&L	THB -493,979.27

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 13 days, 72% target on 3 days.



Risk to this hypothesis: A pullback can mark a lasting trend reversal. This score has no guarantee of mean reversion.

Largest closing position: 9.39%. Maximum measured daily replay time: 28.86s. Timing is local and includes concurrent experiments; it is not a production guarantee.

C5 · Balanced ensemble

CUSTOM STRATEGY

Combine the cross-sectional ranks of the four custom signals equally. This reduces dependence on a single scoring formula, while keeping the decision explainable.

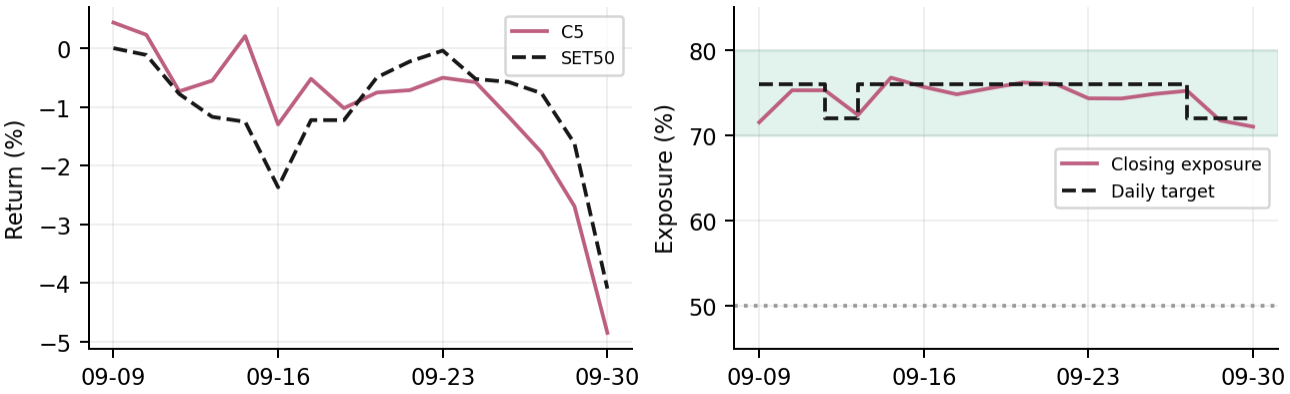
Score: Mean percentile rank of C1, C2, C3 and C4

Inputs: All custom features; average ranks for ties. Daily ranks use the frozen ten-stock basket.

Frozen basket and initial target weights: PTT (9.12%), KBANK (8.78%), PTTGC (8.44%), CPF (8.11%), BCP (7.77%), KKP (7.43%), IVL (7.09%), BDMS (6.76%), SCC (6.42%), KTC (6.08%)

Metric	Value	Metric	Value
Net return	-4.855%	SET50 return	-4.099%
Excess return	-0.756 pp	SET50 MaxDD	4.099%
Official MaxDD	5.268%	Initial-capital MaxDD	5.268%
Average exposure	74.396%	Final exposure	70.977%
Distinct stocks	10	Completed FIFO lots	39
Expected payoff	THB -825.33	Fees	THB 19,214.41
Realized P&L	THB -47,213.84	Unrealized P&L	THB -438,290.57

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 13 days, 72% target on 3 days.



Risk to this hypothesis: The components overlap and can fail together; averaging ranks can dilute a useful signal.

Largest closing position: 9.35%. Maximum measured daily replay time: 32.25s. Timing is local and includes concurrent experiments; it is not a production guarantee.

T1 · SMA trend

INDICATOR STRATEGY

Favor a 20-session simple moving average above the 60-session average, scaled by volatility. This is a slower trend allocation.

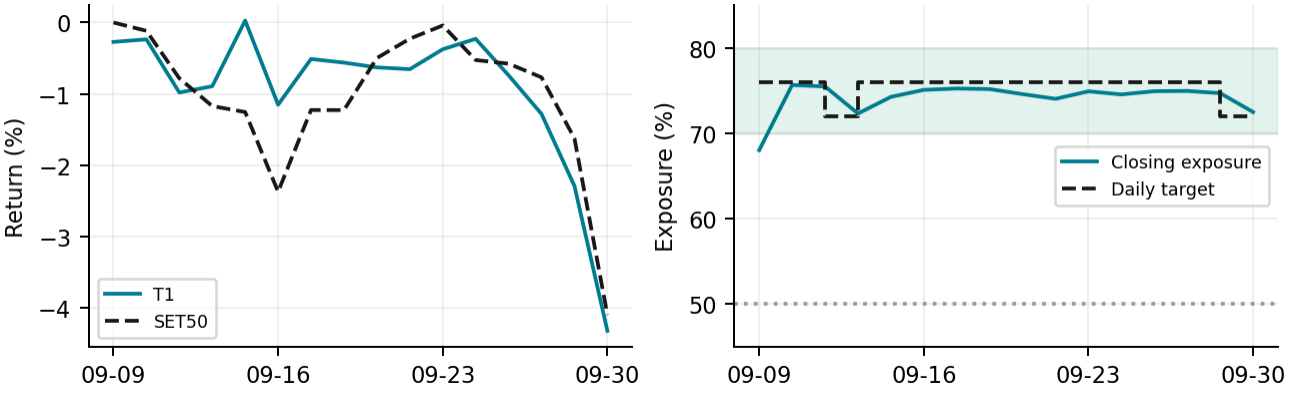
Score: (SMA20 / SMA60 − 1) / volatility20

Inputs: 20- and 60-close simple moving averages; 20-session daily return volatility.

Frozen basket and initial target weights: BCP (9.12%), KBANK (8.78%), PTT (8.44%), PTTGC (8.11%), CPF (7.77%), TTB (7.43%), CRC (7.09%), KTC (6.76%), BH (6.42%), BDMS (6.08%)

Metric	Value	Metric	Value
Net return	-4.324%	SET50 return	-4.099%
Excess return	-0.224 pp	SET50 MaxDD	4.099%
Official MaxDD	4.350%	Initial-capital MaxDD	4.350%
Average exposure	74.123%	Final exposure	72.458%
Distinct stocks	10	Completed FIFO lots	17
Expected payoff	THB -472.03	Fees	THB 15,515.93
Realized P&L	THB -13,124.49	Unrealized P&L	THB -419,233.44

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 14 days, 72% target on 2 days.



Risk to this hypothesis: Moving averages lag turning points and can repeatedly change ranks in sideways markets.

Largest closing position: 9.36%. Maximum measured daily replay time: 30.42s. Timing is local and includes concurrent experiments; it is not a production guarantee.

T2 · EMA trend

INDICATOR STRATEGY

Use faster exponential averages to emphasize recent price information. Allocate more to a stronger normalized separation between the two averages.

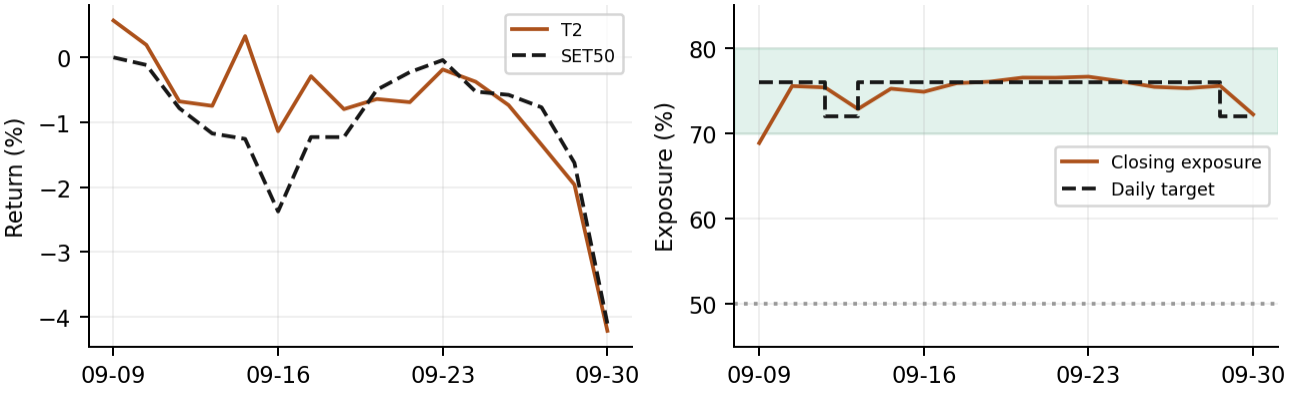
Score: (EMA12 − EMA26) / (close × volatility20)

Inputs: 12- and 26-session EMAs; 20-session volatility. EMA alpha = 2 / (period + 1).

Frozen basket and initial target weights: BCP (9.12%), PTTGC (8.78%), PTT (8.44%), IVL (8.11%), KKP (7.77%), CPF (7.43%), KBANK (7.09%), CCET (6.76%), BDMS (6.42%), CRC (6.08%)

Metric	Value	Metric	Value
Net return	-4.213%	SET50 return	-4.099%
Excess return	-0.113 pp	SET50 MaxDD	4.099%
Official MaxDD	4.754%	Initial-capital MaxDD	4.754%
Average exposure	74.899%	Final exposure	72.182%
Distinct stocks	10	Completed FIFO lots	25
Expected payoff	THB -393.96	Fees	THB 16,227.38
Realized P&L	THB -21,927.17	Unrealized P&L	THB -399,355.21

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 14 days, 72% target on 2 days.



Risk to this hypothesis: Faster reactions can produce more turnover and false signals after brief moves.

Largest closing position: 9.61%. Maximum measured daily replay time: 35.48s. Timing is local and includes concurrent experiments; it is not a production guarantee.

T3 · MACD acceleration

INDICATOR STRATEGY

Combine MACD acceleration with a smaller contribution from the underlying trend. A rising histogram can identify improving momentum before a slow crossover.

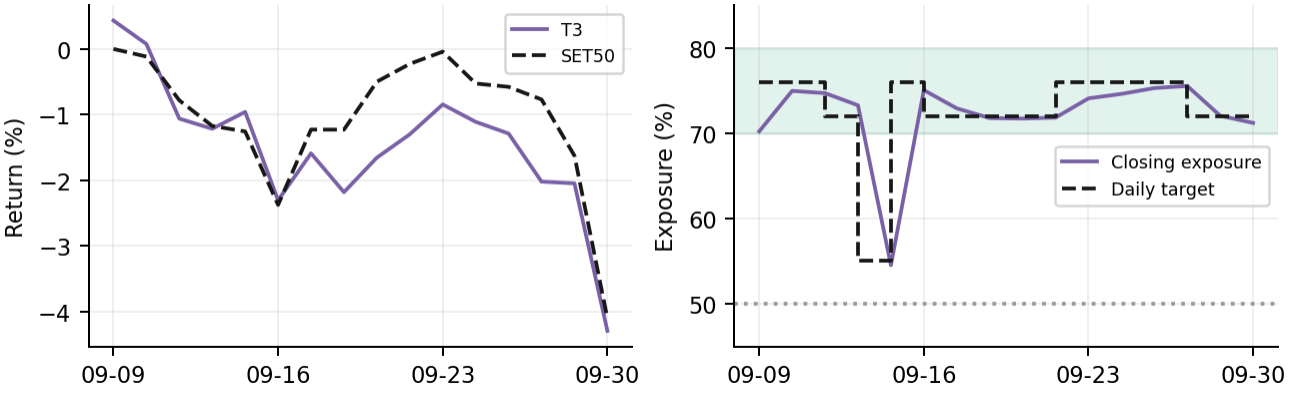
Score: (MACD histogram + 0.25 × MACD line) / (close × volatility20)

Inputs: MACD line = EMA12 − EMA26; signal = EMA9 of MACD; histogram = line − signal.

Frozen basket and initial target weights: IVL (9.12%), PTTGC (8.78%), BCP (8.44%), PTT (8.11%), CCET (7.77%), BDMS (7.43%), DELTA (7.09%), KKP (6.76%), SCC (6.42%), BBL (6.08%)

Metric	Value	Metric	Value
Net return	-4.290%	SET50 return	-4.099%
Excess return	-0.190 pp	SET50 MaxDD	4.099%
Official MaxDD	4.701%	Initial-capital MaxDD	4.701%
Average exposure	72.087%	Final exposure	71.185%
Distinct stocks	10	Completed FIFO lots	61
Expected payoff	THB -618.61	Fees	THB 23,953.53
Realized P&L	THB -71,046.99	Unrealized P&L	THB -357,941.55

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 55% target on 1 day, 76% target on 8 days, 72% target on 7 days.



Risk to this hypothesis: Acceleration can improve while prices still fall. Histogram changes are not proof of a reversal.

Largest closing position: 10.11%. Maximum measured daily replay time: 29.88s. Timing is local and includes concurrent experiments; it is not a production guarantee.

T4 · RSI pullback

INDICATOR STRATEGY

Prefer RSI near 35, with an additional preference for stocks above their 60-session average. This seeks moderate oversold entries within a stronger trend.

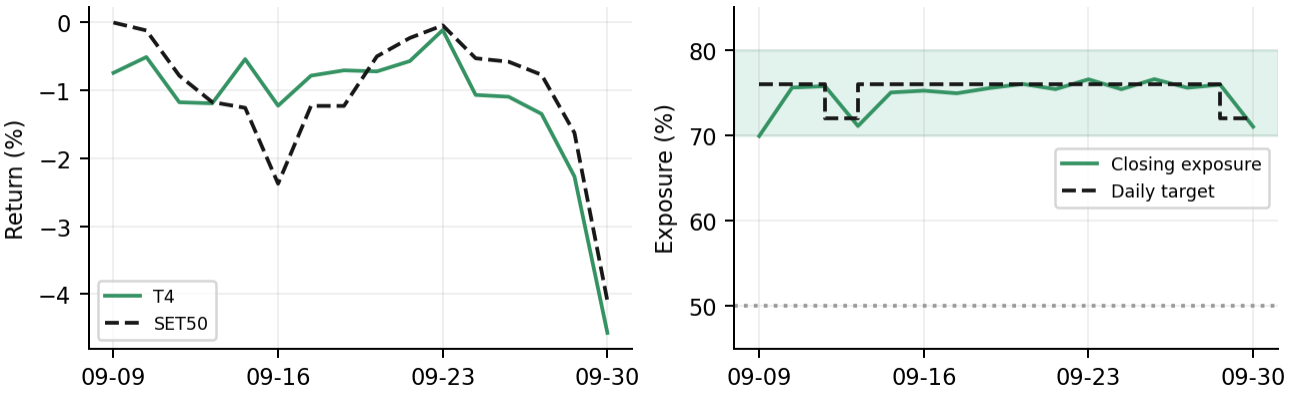
Score: −(RSI14 − 35)² / 100 + (5 if close > SMA60, else 0)

Inputs: Wilder-smoothed 14-session RSI and 60-close simple moving average.

Frozen basket and initial target weights: GPSC (9.12%), TTB (8.78%), KTC (8.44%), KBANK (8.11%), BDMS (7.77%), MTC (7.43%), BH (7.09%), ADVANC (6.76%), CPN (6.42%), THAI (6.08%)

Metric	Value	Metric	Value
Net return	-4.570%	SET50 return	-4.099%
Excess return	-0.470 pp	SET50 MaxDD	4.099%
Official MaxDD	4.464%	Initial-capital MaxDD	4.570%
Average exposure	74.689%	Final exposure	70.974%
Distinct stocks	10	Completed FIFO lots	54
Expected payoff	THB -654.89	Fees	THB 26,261.28
Realized P&L	THB -87,163.07	Unrealized P&L	THB -369,827.21

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 76% target on 14 days, 72% target on 2 days.



Risk to this hypothesis: Oversold stocks can stay oversold; the trend bonus is a preference, not a mandatory filter.

Largest closing position: 9.35%. Maximum measured daily replay time: 43.21s. Timing is local and includes concurrent experiments; it is not a production guarantee.

T5 · Bollinger reversion

INDICATOR STRATEGY

Prefer a close around one standard deviation below its 20-session mean. Penalize falls beyond two deviations and favor prices above the longer average.

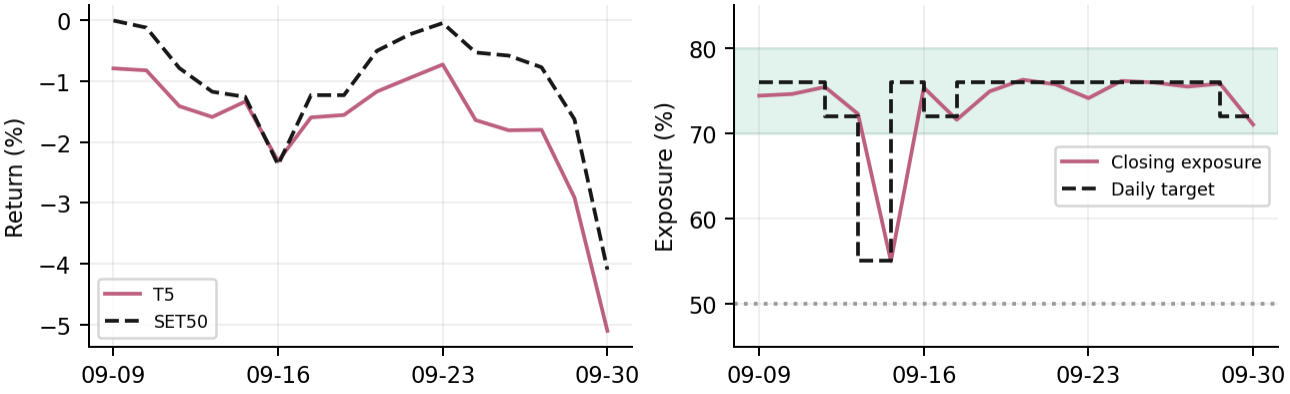
Score: −abs(z20 + 1) − 2 × max(−2 − z20, 0) + (1 if close > SMA60, else 0)

Inputs: z20 = (close − SMA20) / sample standard deviation of 20 closes; SMA60 trend preference.

Frozen basket and initial target weights: KBANK (9.12%), GPSC (8.78%), TTB (8.44%), KTC (8.11%), ADVANC (7.77%), AOT (7.43%), TRUE (7.09%), MTC (6.76%), THAI (6.42%), GULF (6.08%)

Metric	Value	Metric	Value
Net return	-5.107%	SET50 return	-4.099%
Excess return	-1.008 pp	SET50 MaxDD	4.099%
Official MaxDD	4.416%	Initial-capital MaxDD	5.107%
Average exposure	73.351%	Final exposure	70.991%
Distinct stocks	10	Completed FIFO lots	72
Expected payoff	THB -2,221.38	Fees	THB 33,162.82
Realized P&L	THB -172,507.93	Unrealized P&L	THB -338,197.89

Common: fail; Best Alpha: fail; user exposure band: met. Unmet common criteria: positive expected payoff. Target days: 55% target on 1 day, 76% target on 12 days, 72% target on 3 days.



Risk to this hypothesis: Price distributions need not be normal, and lower-band moves can become persistent declines.

Largest closing position: 9.27%. Maximum measured daily replay time: 35.14s. Timing is local and includes concurrent experiments; it is not a production guarantee.

How to read these measurements

Payoff and expected payoff

A completed trade’s payoff is its net profit or loss after both-side costs. Here a trade means a fully closed FIFO buy lot. TradeSim expected payoff is the arithmetic mean P&L of completed non-breakeven lots: win fraction × mean win − loss fraction × mean loss magnitude. Six wins of THB 1,000 and four losses of THB 800 imply THB 280 per completed lot. It is an observed sample average, not a forecast. Partial closes of still-open lots affect realized P&L but not completed-lot payoff yet; zero completed non-flat lots means undefined.

Simulation data

Yes: executions and valuation use the stock ticks supplied with TradeSim. Prior stock/SET50 daily history provides signals; the supplied SET50 benchmark file provides benchmark metrics. Each strategy gets the same dates, capital, costs and input package. Its selected basket and resulting trades differ. No synthetic prices are used for reported returns.

Percent versus percentage points

A strategy return of 3% minus a benchmark return of 1% is 2 percentage points (2 pp) of excess return. Both inputs are percentages; their arithmetic difference is reported in pp. This is not the compounded wealth ratio (1.03 / 1.01 − 1).

Return and drawdown anchors

Net return is final NAV / initial THB 10 million − 1. SET50 return uses the first sample close. Official maximum drawdown uses daily closing peaks and therefore excludes an initial-capital peak before the first close. Initial-capital drawdown and aligned first-close returns are also saved separately. Intraday drawdown is not this official metric.

Exposure and counts

Average exposure is the mean of the 16 daily closing invested-value / NAV fractions, including the first day. It is not the target, final value or time-weighted intraday exposure. Distinct stocks counts unique names with fills. Completed FIFO lots can be numerous because IOC orders create many small lots; their count is not a count of independent investment ideas.

Prize conditions

Working common criteria: portfolio MaxDD strictly below 1.5× SET50 MaxDD, at least five completed FIFO lots, average exposure at least 50%, and expected payoff above zero. At least five distinct stocks is tracked separately. Positive total return is not a universal common requirement. Best Alpha additionally requires nonnegative total and excess returns. Passing these local checks is not organizer acceptance or a prize guarantee.

Evidence, limitations and reproduction

61 automated tests pass, including the original 17 TradeSim contract tests and indicator mathematics checked independently against pandas. Daily accounting reconciles for all 160 strategy-days. Repeatability: pass. C1 was replayed in a fresh process; all 9 deterministic artifacts match byte for byte. The original Phase 2 sources and pinned package remain unchanged. Full run bundles stay under ignored artifacts/; compact metrics, daily series and hashes are committed with this report.

Only 16 trading days are available for tick-level evaluation, and this sample has been reused. The supplied 48-stock universe is static; historical membership and price-adjustment semantics are not independently established. Cross-symbol marks are not synchronized under notebook event ordering. Closing drawdown omits intraday extremes. No robustness, out-of-sample, production acceptance or future-return claim is made. A ten-stock basket limits diversification and replacement choices; expanding it belongs to a later declared experiment.

Reproduction

uv pip install --python .venv/bin/python -r requirements-reporting.txt

.venv/bin/python -m unittest discover -s tests -v

.venv/bin/python scripts/run_strategy_suite.py --output artifacts/comparison --workers 3

.venv/bin/python scripts/run_phase3.py --config research/phase3-suite/C1.json --output artifacts/repeat

.venv/bin/python scripts/build_phase3_results.py --input artifacts/comparison --repeat artifacts/repeat

First install the pinned Python/TradeSim environment using README.md. To independently regenerate baskets, use scripts/prepare_strategy_suite.py --output artifacts/prepared-baskets and compare its manifest/configs with research/phase3-suite/. New run directories must not already exist.

Phase status

Phase 3 meets the development guide’s engineering checkpoint: complete replay, reconciled accounts, compatible single-file strategy and reproducible results. The ten hypotheses also establish the comparison structure for Phase 4. Phase 4 is not complete: additional permitted data and chronological validation are still required.

Current achievements: docs/current-progress/README.md. Next actions and lessons: docs/to-do/README.md. Working rules and source limits: docs/award-criteria.md. Exact predeclared formulas: docs/phase-3-ten-strategy-spec.md.